



Week 6

- **Design Principles**
- **Software Engineering with AI**

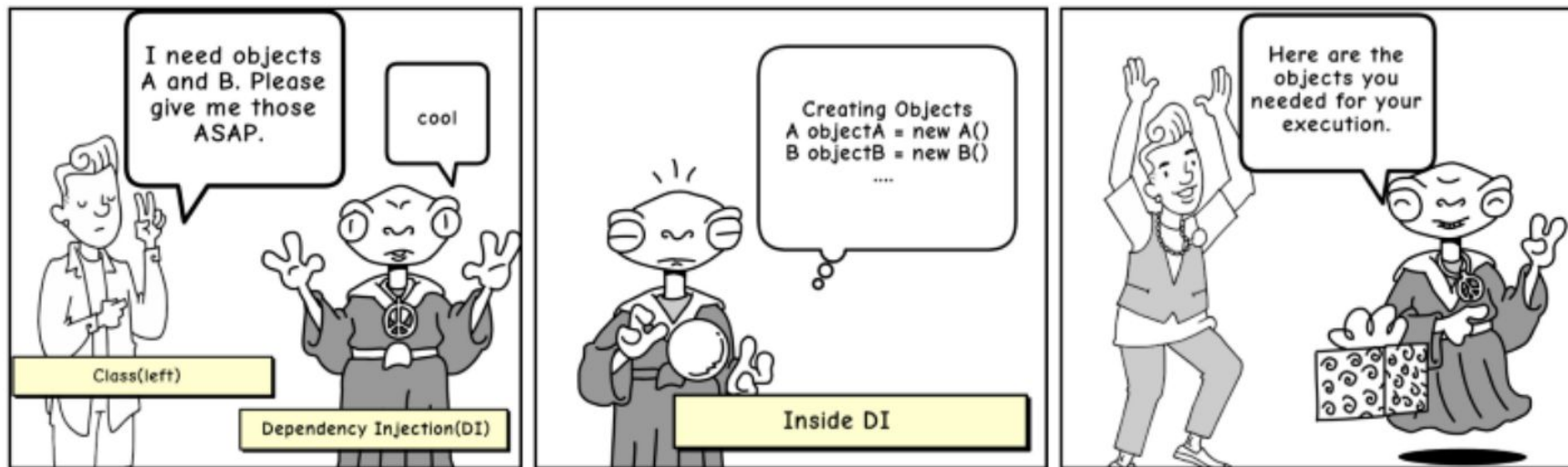
Dependency Injection

Dependency Injection is a technique whereby one object (or static method) supplies the dependencies of another object. A dependency is an object that can be used (a service) - wikipedia

When class A uses some functionality of class B, then its said that class A has a dependency of class B.

In Java, before we can use methods of other classes, we first need to create the object of that class (i.e. class A needs to create an instance of class B).

So, transferring the task of creating the object to someone else and directly using the dependency is called dependency injection.



This comic was created at www.MakeBeliefsComix.com. Go there and make one now!

What if code could speak?

ProductCatalog

ProductRepository

SQLProductRepository

```
import java.util.List;

public class ProductCatalog {
    private ProductRepository productRepository;

    public ProductCatalog(ProductRepository productRepository) {
        this.productRepository = productRepository;
    }

    public void listAllProducts() {
        List<String> allProductNames =
            productRepository.getAllProductNames();

        // Display product names
    }
}
```

```
import java.util.List;

public interface ProductRepository {

    public List<String> getAllProductNames();

}
```

```
import java.util.Arrays;
import java.util.List;

public class SQLProductRepository implements ProductRepository{

    public List<String> getAllProductNames() {
        return Arrays.asList("soap", "toothpaste");
    }

}
```

```
public class ECommerceMainApplication {

    public static void main(String[] args) {

        ProductRepository productRepository =
            ProductFactory.create();

        ProductCatalog productCatalog = new
            ProductCatalog(productRepository);

        productCatalog.listAllProducts();

    }

}
```

```
public class ProductFactory {

    public static ProductRepository create() {
        return new SQLProductRepository();
    }

}
```



Gen AI vs Traditional AI

Traditional AI

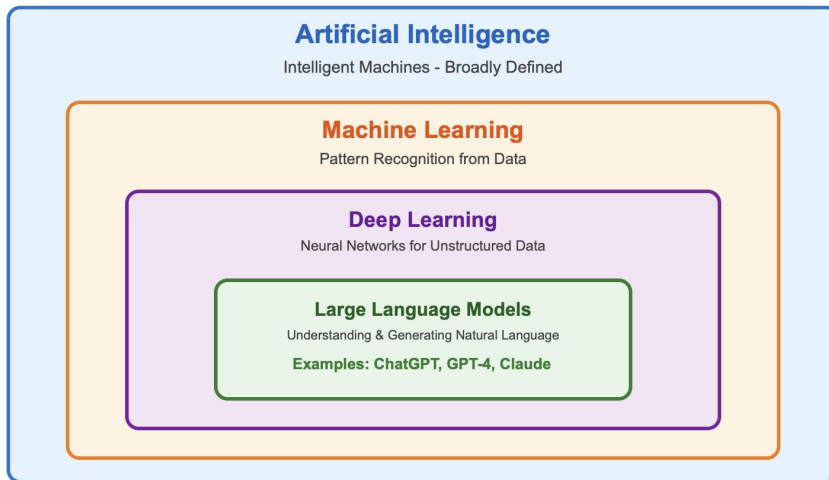
- Focuses on classification, prediction , and automation
- Learns from labeled data using rules or patterns
- Works within set boundaries (if X happens, do Y)
- Examples - Spam filters, voice assistant, fraud detection etc

Gen AI

- Focuses on creating new pattern from learned pattern.
- Generate text, code, images based on prompts
- Examples, chatbot like chatGPT, Claude etc.



- **Artificial Intelligence (AI)** is very a broad term, but generally it deals with intelligent machines.
- **Machine Learning (ML)** is a subfield of AI that specifically focuses on pattern recognition in data. As you can imagine, once you recognize a pattern, you can apply that pattern to new observations. That's the essence of the idea, but we will get to that in just a bit.
- **Deep Learning** is the field within ML that is focused on unstructured data, which includes text and images. It relies on artificial neural networks, a method that is (loosely) inspired by the human brain.
- **Large Language Models (LLMs)** deal with text specifically





What are LLM

A Large Language Model, or LLM is a type of AI trained to understand and generate human like text based on patterns in massive amount of language data

- Trained on billions of words from book, websites, articles and conversations
- Learns grammar, meaning, context, and relationship between words
- Can answer questions, translate text, write content, generate new content
- Not thinking like a human, but using patterns it learned during training.



Machine Learning

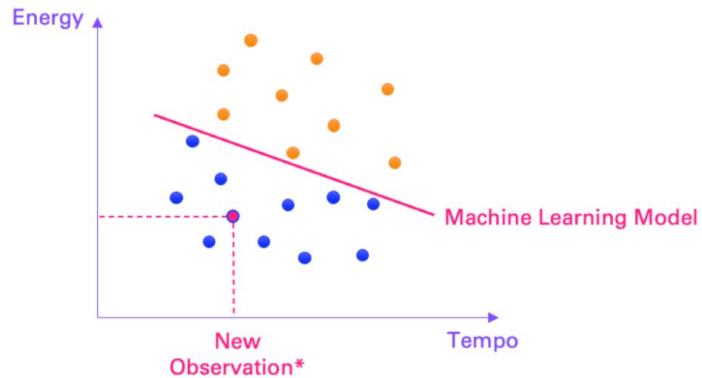
Machine Learning is to discover patterns in data. Or more specifically, a pattern that describes the relationship between an input and an outcome.

Example - Let's say we would like to distinguish between two genres of music: *reggaeton* and *R&B*.

Reggaeton is a Latin urban genre known for its lively beats and danceable rhythms, while R&B (Rhythm and Blues) is a genre rooted in African-American musical traditions, characterized by soulful vocals and a mix of upbeat and slower-paced songs

Classification Example: Predicting Music Genre

● R&B Songs ● Reggaeton Songs ● New Observation



*New Observation:

[**Tempo: 30, Energy: 20**]

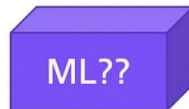


Class	Probability
R&B	0.92
Reggaeton	0.08

If the input is Image

Classification

Is it a **tiger**, a **cat**, or a **fox**?

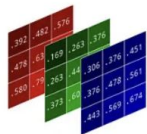


Class	Probability
Dog	0.03
Cat	0.96
Bird	0.01

=



=



=

224x224x3 = **150,528** pixels (!!)

If the Input is text

[The movie was great]

0.4	-0.2	0.3	0.5
-0.1	0.6	-0.2	0.9
0.8	-0.3	0.4	-0.7
...	...	...	...
0.2	-0.7	-0.6	-0.3

ML??

Class	Probability
Positive	0.98
Negative	0.02

Word **embeddings** represent a word's meaning in its context.



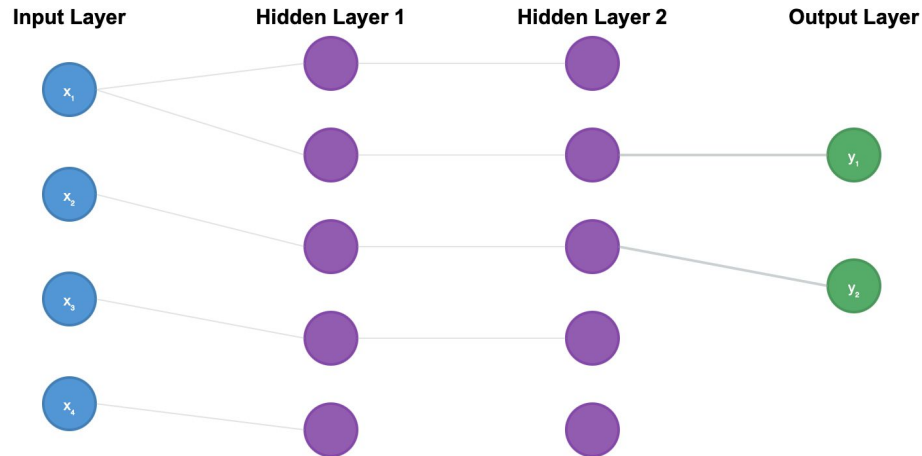
Neural Networks

Neural networks are powerful Machine Learning models that allow arbitrarily complex relationships to be modeled.

They are the engine that enables learning such complex relationships at massive scale.

- They consist of a sequence of layers of connected “neurons” that an input signal passes through in order to predict the outcome variable.
- Multiple layers of linear regression stacked together, with the addition of non-linearities in between, which allows the neural network to model highly non-linear relationships

A simplified neural network showing how information flows from input through hidden layers to output, with each connection representing a learnable parameter. For Ex. ChatGPT, for example, is based on a neural network consisting of 176 billion neurons, which is more than the approximate 100 billion neurons in a human brain.



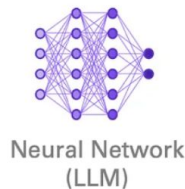
Imagine the following task: Predict the next word in a sequence

[The cat likes to sleep in the _____] → What **word** comes next?

Can we frame this as a ML problem? Yes, it's a **classification** task.

Now we have (say)
~50,000 **classes** (i.e.
words)

[The cat likes to sleep in the]
Input

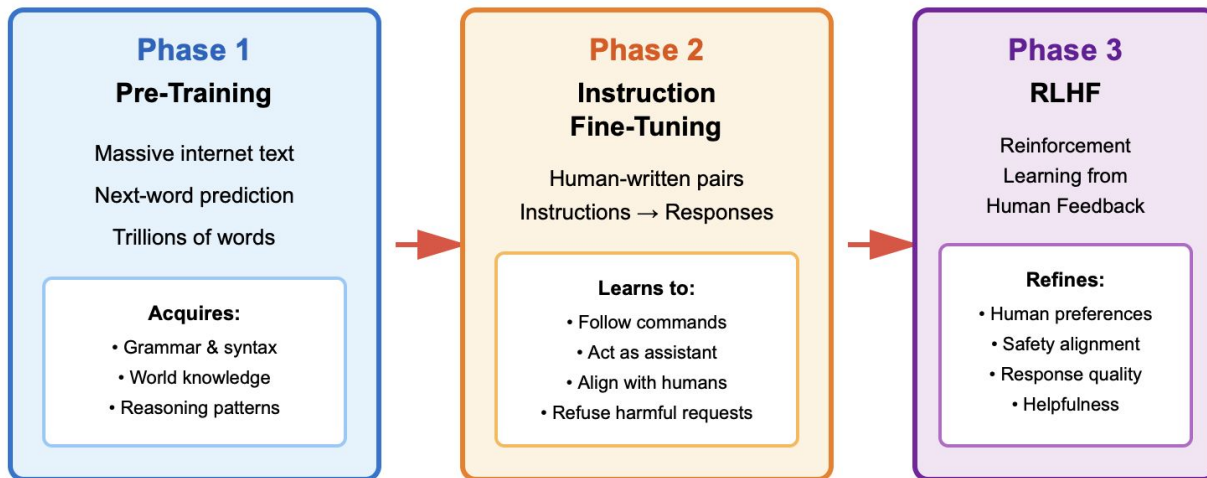


Word	Probability
ability	0.002
bag	0.071
box	0.085
...	...
zebra	0.001

Output

Three Phases of LLM

Modern LLMs undergo multiple training stages to develop both language understanding and helpful assistant behavior.





Tokens

A token is a piece of text - it could be word, part of word, or even punctuation, or space depending on how text is split OR token is like a building block of language that a model understands

<https://gptforwork.com/tools/tokenizer>



LLM As Knowledge base

By obtaining the knowledge from the training data, return useful information for the user.

Example: A UTSC Registrar chat bot



Gemini

Misunderstood the user's request/ didn't differentiate between CSCC01 and CSCD01

Who is the current instructor of CSCC01 at UTSC?



Based on the information available, the current instructors for CSCC01 (Introduction to Software Engineering) at UTSC are **Cho Yin Yong** and **Aleksander Bodurri**. They are listed for both LEC01 and LEC02 sections.

“Hallucinations” can have bad side effects



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Airline held liable for its chatbot giving passenger bad advice - what this means for travellers

23 February 2024

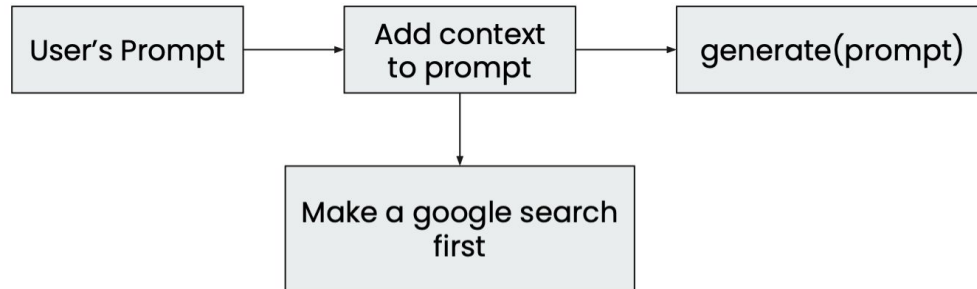
Share Save

should pay. In 2022, Air Canada's chatbot promised a discount that wasn't available to passenger Jake Moffatt, who was assured that he could book a full-fare flight for his grandmother's funeral and then apply for a bereavement fare after the fact.

LLM As a Logical orchestrator - RAG

Retrieving relevant information prior to Generative AI inference.

Example: Make a google search before any AI inference to populate the prompt with the the right context.



Text in text out - same interface - easy win

Claude - Outdated Data

CY Who is the current instructor of CSCC01 at UTSC?

🌐 CSCC01 instructor UTSC 2025

10 results ▾

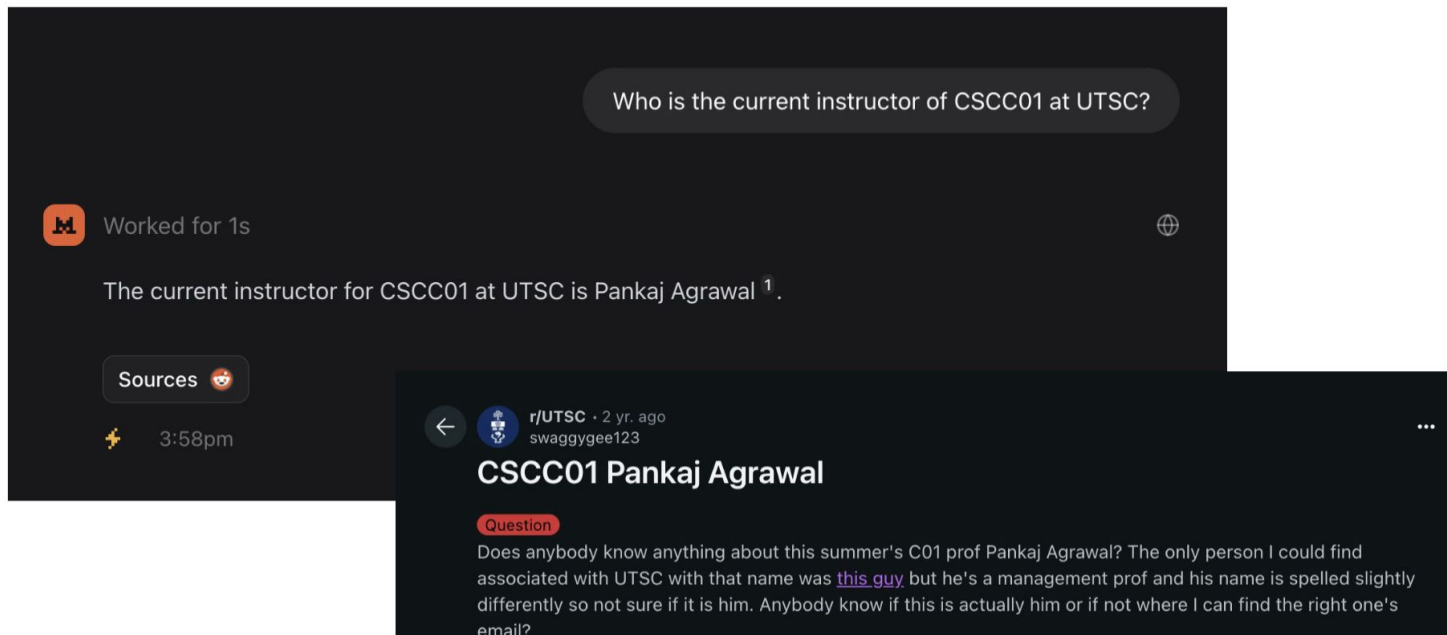
I can see some information about CSCC01 at UTSC, but I need to get more specific information about the current instructor. Let me search for more recent information.

🌐 "Thierry Sans" CSCC01 UTSC 2025

7 results ▾

Based on my search results, Thierry Sans appears to be the current instructor of CSCC01 at UTSC [Rate My Professors](#) [GitHub](#) . He is an Assistant Professor, Teaching Stream in the Department of Computer and Mathematical Sciences at University of Toronto Scarborough [ThierrySans \(Thierry Sans\)...](#) , and the course website shows his name associated with CSCC01 Introduction to Software Engineering [Thierry Sans at University ...](#) .

Mistral - data from non-authoritative sources



Who is the current instructor of CSCC01 at UTSC?

Worked for 1s

The current instructor for CSCC01 at UTSC is Pankaj Agrawal ¹.

Sources

3:58pm

r/UTSC · 2 yr. ago
swaggygee123

CSCC01 Pankaj Agrawal

Question

Does anybody know anything about this summer's C01 prof Pankaj Agrawal? The only person I could find associated with UTSC with that name was [this guy](#) but he's a management prof and his name is spelled slightly differently so not sure if it is him. Anybody know if this is actually him or if not where I can find the right one's email?

ChatGPT - correct!

Who is the current instructor of CSCC01 at UTSC?

The current instructor for **CSCC01 – Introduction to Software Engineering** at UTSC (Summer 2025) is **Pankaj Agrawal**, as listed on the course's official Piazza page [piazza.com +9](#) .

If you're looking for details like office hours, lecture schedule, or TA assignments, I can help you dig deeper—just let me know!

        Sources



Function Calling

- Create deterministic functions that takes in specific input.
 - a. `get_instructor(course_code: str) -> str`
 - SQL search
- Instead of relying on language model to give direct output, ask it to select the most appropriate function
- We are utilizing the LLM to:
 - Understand the user's intent and select the right function
 - Successfully extract the course code from the user's query

Function calling example

You are an assistant that has access to the following set of tools.
Here are the names and descriptions for each tool:

`get_instructor(course_code: str) -> str`: Given a course code, return the instructor
`get_courses(instructor_name: str) -> str`: Given an instructor name, return courses taught

Given the user input, return the name and input of the tool to use.
Return your response as a JSON blob with 'name' and 'arguments' keys.

The `arguments` should be a dictionary, with keys corresponding to the argument names and the values corresponding to the requested values.

Who is the instructor of CSCC01?

```
{"name": "get_instructor", "arguments": {"course_code": "CSCC01"}}
```

Who is the instructor of CSCC01? Pankaj Agrawal

The instructor of CSCC01 is Pankaj Agrawal.



Content Generation

Use LLM to generate large chunks of content, based on an initial set of instructions/constraints as described in the input prompt.

- Essay generation
- Code generation
 - Github Copilot - simple code autocompletion
 - Github Copilot Chat - code editor
 - Lovable - full app creation

Github Copilot - Code autocompletion

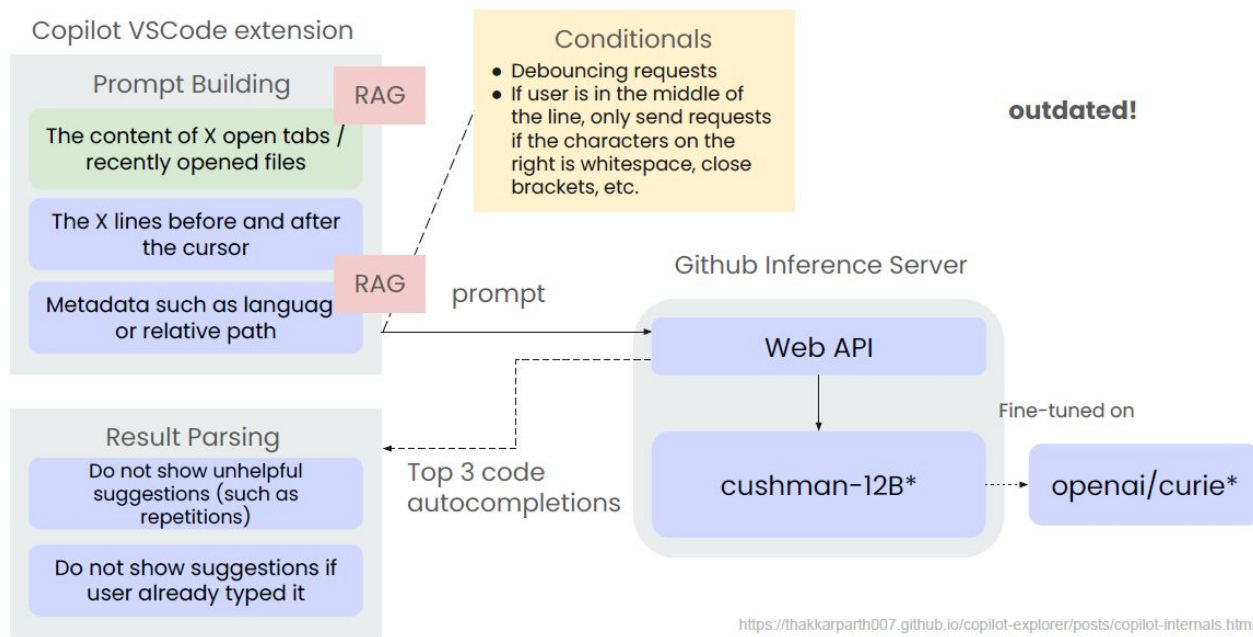
The user's prompt is constructed by the Copilot extension.

Possible Components of the prompt:

- N characters before the cursor
- N characters after the cursor
- What else?

```
searchRoutes.openapi(getReferrer, async (c) => {  
  const referrerId = c.req.param("referrerId");  
  if (!referrerId) {  
    return c.json({ error: "Referrer ID is required" }, 400);  
  }  
  const user = (await c.db.select({ name: users.name, id: users.id }).from(users)  
  if (!user) {  
    return c.json({ error: "Referrer not found" }, 404);  
  }  
})
```

Reverse Engineered Github Copilot Architecture (2024)





How I use Github Copilot

Write a comment + initial code and hope that it autocompletes it
in the next line.

```
// obtain user object
```

```
const user =
```

Success rate of ~60%!



How I measure the “success rate”

The autocompletion should be exactly what I expect had I wrote the code myself.

Opinion: LLMs can make me type lazier but it does not release my brain from being in control.

Github Copilot Chat

Give it a file / multiple files, and it will make the changes for you.

What is our role in this process?

“Code review”

Opinion: LLMs can make me type lazier but it does not release my brain from being in control.

Apple now supports direct link installation

if the user's device is an apple iphone then show a button that links to https://esimsetup.apple.com/esim_qrcode_provisioning?carddata={activationCode}

The button should be apple styled "Add eSIM to phone"

QrCodeSection.tsx:9-27

GitHub Copilot

> Used 1 reference

QrCodeSection.tsx

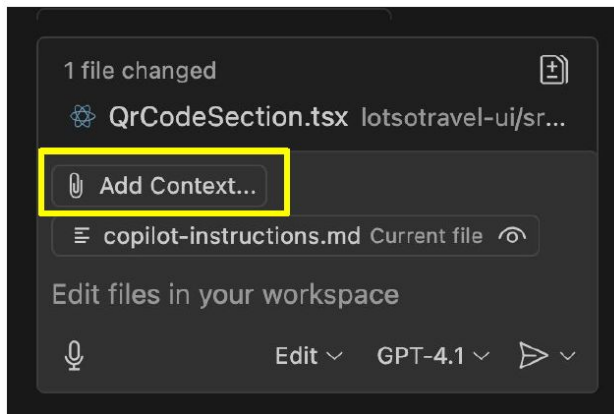
Add Apple eSIM direct install button if the user's device is an iPhone. The button links to the Apple eSIM provisioning URL and is styled in an Apple-like way.

1 file changed

QrCodeSection.tsx lotsotravel-ui/sr...

Tip 1: Select the right files as context

You could rely on the built-in RAG capabilities, but, you can also tell the LLM which files to look at manually.





Tip 2: Build custom prompts to give LLM context

For Github Copilot, you can put a file in

`.github/copilot-instructions.md` for more context.

For example, in my codebase, there are example snippets of code of how to use

Cloudflare Workers.

Example Custom Prompt - What to include

```
github > prompts > E game.prompt.md
1 This application uses a modular game architecture where each game mode consists of three main components: TV Display (for the main
  screen), Host Controls (for game management), and Mobile Controller (for player interaction). To add a new game, you'll need to
  create these components and register the game in the system.
2
3 Required Files to Create/Modify
4 1. Create Game Directory Structure
5 src/games/[game_name]/
6 | [GameName]TVDisplay.tsx
7 | [GameName]HostControls.tsx
8 | [GameName]MobileController.tsx
9 | use[GameName]Game.ts
10
11 TV Display Component ([GameName]TVDisplay.tsx)
12 Purpose: Main screen display visible to all players
13 Props: GameTVDisplayProps from ../shared/GameInterface
14 Requirements:
15 Full-screen responsive design
16 Display current game state
17 Show player information (names, avatars, teams/roles if applicable)
18 Use PlayerGrid component for displaying players
19 Example structure:
20
21 import { GameTVDisplayProps } from '../shared/GameInterface';
22 export const [GameName]TVDisplay: React.FC<GameTVDisplayProps> = ({ room, players }) => {
23   return (
24     <div className="min-h-screen bg-gradient-to-br from-[color1] to-[color2]">
25       { /* Game title and status */ }
26       { /* Player display using PlayerGrid */ }
27     </div>
28   );
29 };
```

Introduce custom directory structure.

Introduce custom code format for implementation



What is the point of being in Computer Science then?

If AI can write code, why am I still useful? Short answer - because you can read/write actual code.

A non computer scientist prompting the AI to fix minor bugs:

the report tab has been redesigned instead of using the proper structure and framework. the nav is different than the other tactical and strategic tabs, when they should all use the same nav component for continuity, and it doesn't have the desired capabilities like a toggle to dashboard view. the kanban view is also broken now and doesn't allow for drag and drop. find the old kanban board code and use that instead. make sure you stop breaking old functionality, you redesigned the report module of projects instead of adding to it, make sure this doesn't happen again. perform a comprehensive scan of the codebase to correct this effectively

the report tab existed before in the dashboard route, see if its still there first!

doesn't the existing FeedbackViewMode already have a kanban and table view?

resolve this build error properly using our lessons learned on not building new code to replace existing functionality. perform a proper analysis of our ProjectDashboard architecture first so you don't redo functionality we already have



When software engineers prompt AI

Read the API specifications and create a `lib/api.ts` with functions interacting with the API

- Split things into separate components using the following method...
- (Instead of prompting AI to change button colors - just edit the CSS)
- More technical words used, that allows us to be in control of LLM inference!



Notebook LM Demo

<https://notebooklm.google/>



Thank You